# SOP-HX-160: Heat Exchanger Tube Bundle Inspection and Retirement

**Document owner:** Mechanical Integrity Department **Effective date:** 2024-05-20 **Review cycle:** 24 months

## Section 1. Purpose and Scope

Governs eddy-current and visual inspection of shell-and-tube heat exchanger bundles, and the retirement-thickness evaluation triggered by degradation findings.

## Section 2. Inspection Timing

Clause 2.1 — Every tube bundle in a fouling or corrosive service is eddy-current tested at each turnaround, not to exceed 48 months.

Clause 2.2 — A bundle exhibiting a process-side leak (tube-to-shell crossover) requires eddy-current testing of the entire bundle at the next planned outage, regardless of turnaround schedule, and immediate isolation of the affected tube(s) in the interim.

## Section 3. Retirement Thickness

Clause 4.6 — Tube bundle retirement wall thickness must be recalculated following any eddy-current inspection showing degradation exceeding the Clause 4.8 engineering-review threshold.

Clause 4.8 — Eddy-current results showing wall loss exceeding 40% of nominal at any single tube support location, or exceeding 25% general wall loss across more than 10% of tubes, trigger mandatory engineering review before the bundle returns to service.

## Section 4. Plugging Limits

Clause 5.1 — Individual tubes below retirement thickness shall be mechanically plugged, not left in service pending the next inspection.

Clause 5.3 — A bundle with more than 10% of its tubes plugged requires an engineering thermal-performance review to confirm continued service does not compromise the associated process's design duty.

## Section 5. Records

Clause 8.1 — Filed to the Heat Exchanger Integrity File. Retain for the operating life of the unit plus 10 years.